Кладковой Максим 5130904/30002 Вар 47

Кусочно-кубическая интерполяция Эрмита

$f(x)=\sin(\pi x)$; $x\in[-\frac{1}{4};\frac{1}{4}]$; $n=3$; $\frac{df}{dx}=\pi\cos(\pi x)$

№	x	$f(x)$	$\frac{df}{dx}=m$
0	$-\frac{1}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$
1	$-\frac{1}{6}$	$-\frac{1}{2}$	$\frac{\pi\sqrt{3}}{2}$
2	0	0	π
3	$\frac{1}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$

$$h_1=-\frac{1}{6}+\frac{1}{4}=\frac{1}{12}\qquad t=\frac{x+\frac{1}{4}}{\frac{1}{12}}=12x+3\qquad x\in[-\frac{1}{4};-\frac{1}{6}]$$

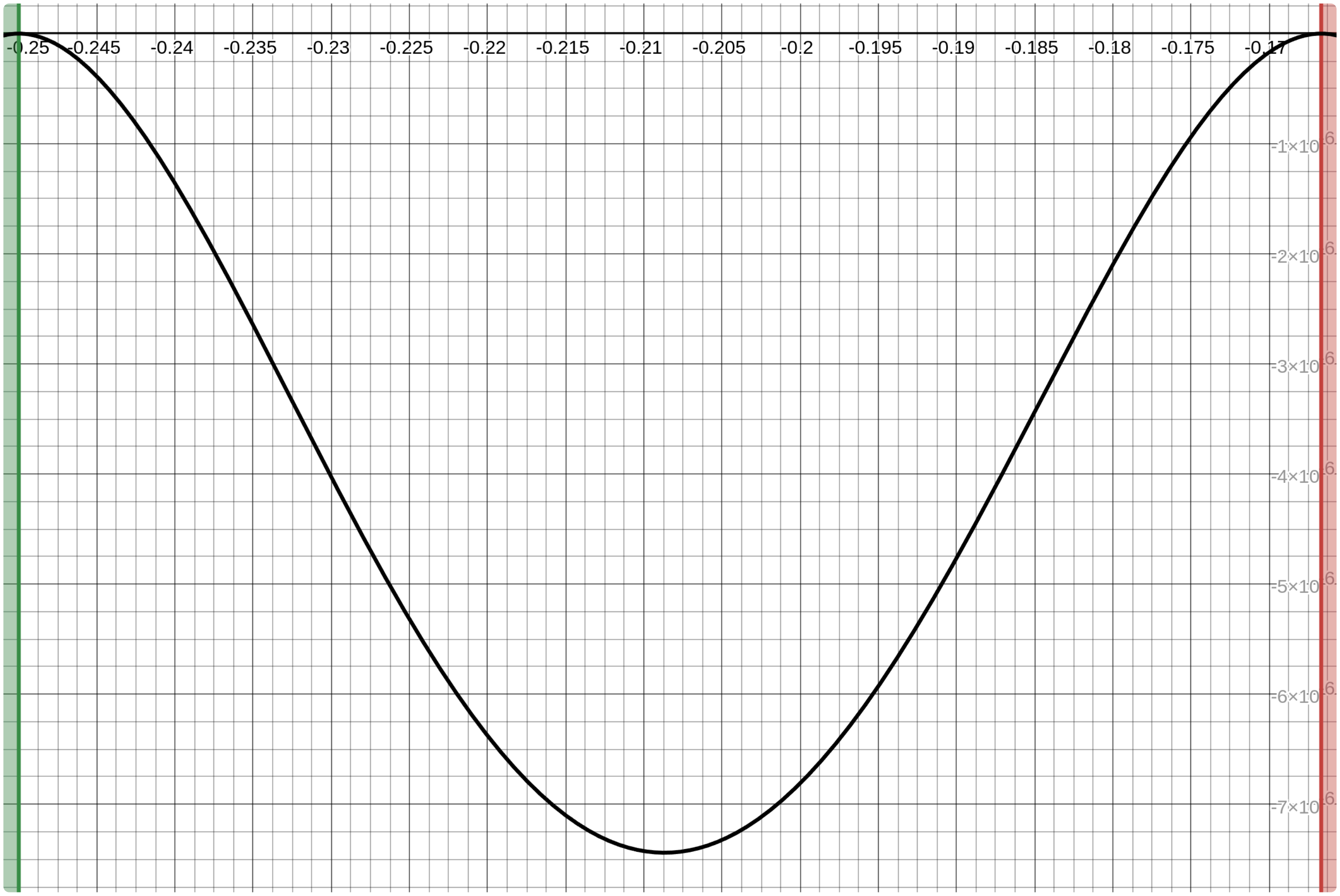
$$H_1(t)=-\frac{\sqrt{2}}{2}(1-3t^2+2t^3)-\frac{1}{2}(3t^2-2t^3)+\frac{\pi\sqrt{2}}{2}\frac{1}{12}(t-2t^2+t^3)+\frac{\pi\sqrt{3}}{2}\frac{1}{12}(-t^2+t^3)$$

$$H_1(x)=-\frac{\sqrt{2}}{2}(1-3(12x+3)^2+2(12x+3)^3)-\frac{1}{2}(3(12x+3)^2-2(12x+3)^3)+\frac{\pi\sqrt{2}}{24}(12x+3-2(12x+3)^2$$
$$+(12x+3)^3)+\frac{\pi\sqrt{3}}{24}(-(12x+3)^2+(12x+3)^3)=-(1728x^3+1080x^2+216x+14)\sqrt{2}+$$
$$+\frac{3456x^3+2160x^2+432x+27}{2}+\frac{\pi(16x+1+84x^2+144x^3)}{\sqrt{2}}+\frac{\pi\sqrt{3}(192x^2+42x+3+288x^3)}{4}$$
$$=x^3(-1728\sqrt{2}+1728+72\pi\sqrt{2}+72\pi\sqrt{3})+x^2(-1080\sqrt{2}+1080+42\pi\sqrt{2}+48\pi\sqrt{3})$$
$$+x(-216\sqrt{2}+216+8\pi\sqrt{2}+\frac{21\pi\sqrt{3}}{2})+(-14\sqrt{2}+\frac{27}{2}+\frac{\pi\sqrt{2}}{2}+\frac{3\pi\sqrt{3}}{4})$$

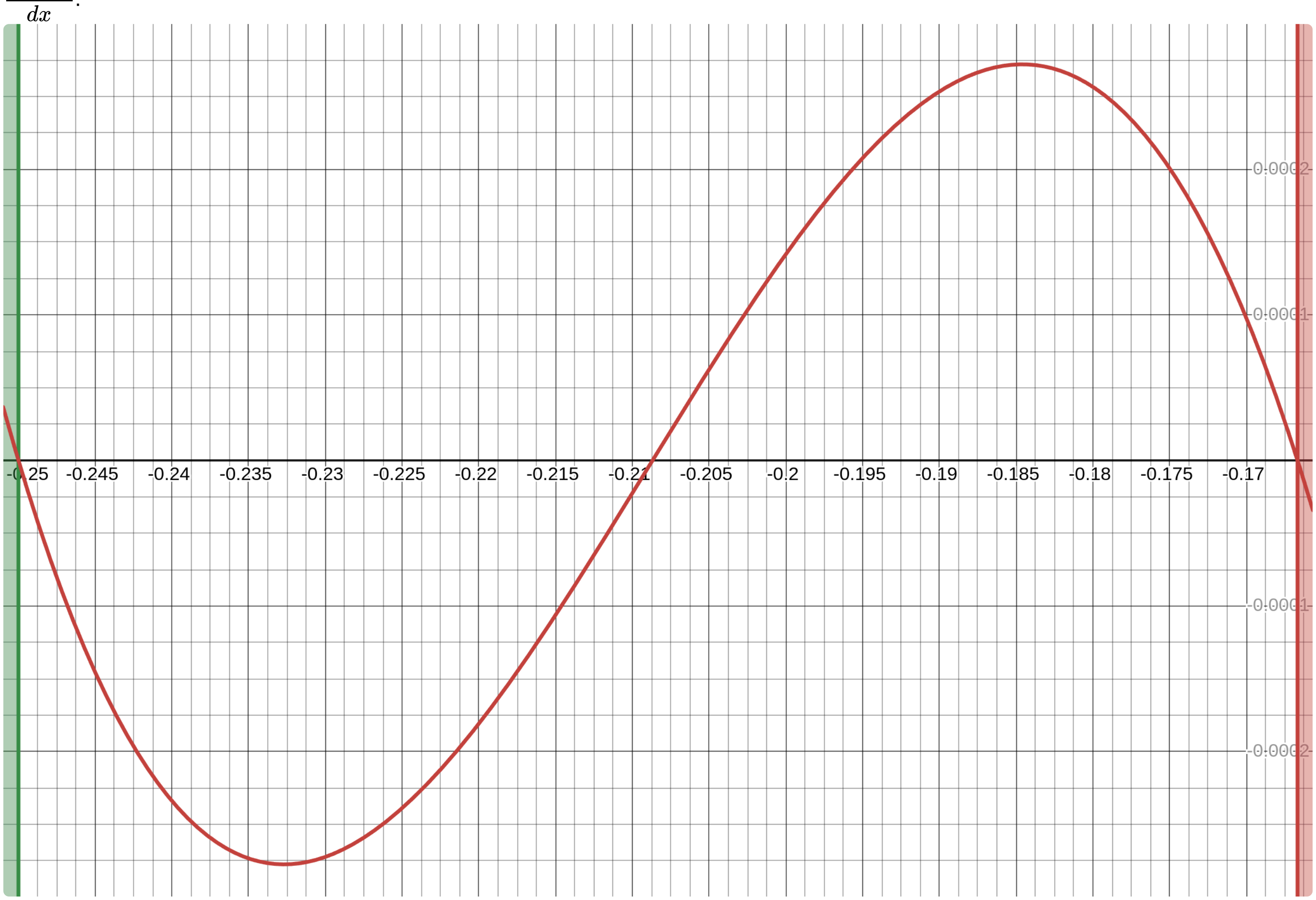
$$r_1(x)=\sin(\pi x)-x^3(-1728\sqrt{2}+1728+72\pi\sqrt{2}+72\pi\sqrt{3})+x^2(-1080\sqrt{2}+1080+42\pi\sqrt{2}+48\pi$$
$$+x(-216\sqrt{2}+216+8\pi\sqrt{2}+\frac{21\pi\sqrt{3}}{2})+(-14\sqrt{2}+\frac{27}{2}+\frac{\pi\sqrt{2}}{2}+\frac{3\pi\sqrt{3}}{4})$$

$$\frac{dr_1(x)}{dx}=\pi\cos(\pi x)-3x^2(-1728\sqrt{2}+1728+72\pi\sqrt{2}+72\pi\sqrt{3})+2x(-1080\sqrt{2}+1080+42\pi\sqrt{2}+48\pi$$
$$+(-216\sqrt{2}+216+8\pi\sqrt{2}+\frac{21\pi\sqrt{3}}{2})+(-14\sqrt{2}+\frac{27}{2}+\frac{\pi\sqrt{2}}{2}+\frac{3\pi\sqrt{3}}{4})$$

$r_1(x)$:



$\frac{dr_1(x)}{dx}$:



$$h_2 = 0 + \frac{1}{6} = \frac{1}{6}; \quad t_2 = \frac{x+\frac{1}{6}}{\frac{1}{6}} = 6x+1 \qquad x \in \left[-\frac{1}{6}; 0\right]$$

$$H_2(t) = -\frac{1}{2}(1-3t^2+2t^3) + 0(3t^2-2t^3) + \frac{\pi\sqrt{3}}{2}\cdot\frac{1}{6}(t-2t^2+t^3) + \frac{\pi}{6}(-t^2+t^3) =$$

$$= \frac{-(1-3t^2+2t^3)}{2} + \frac{\pi\sqrt{3}(t-2t^2+t^3)}{12} + \frac{\pi(-t^2+t^3)}{6}$$

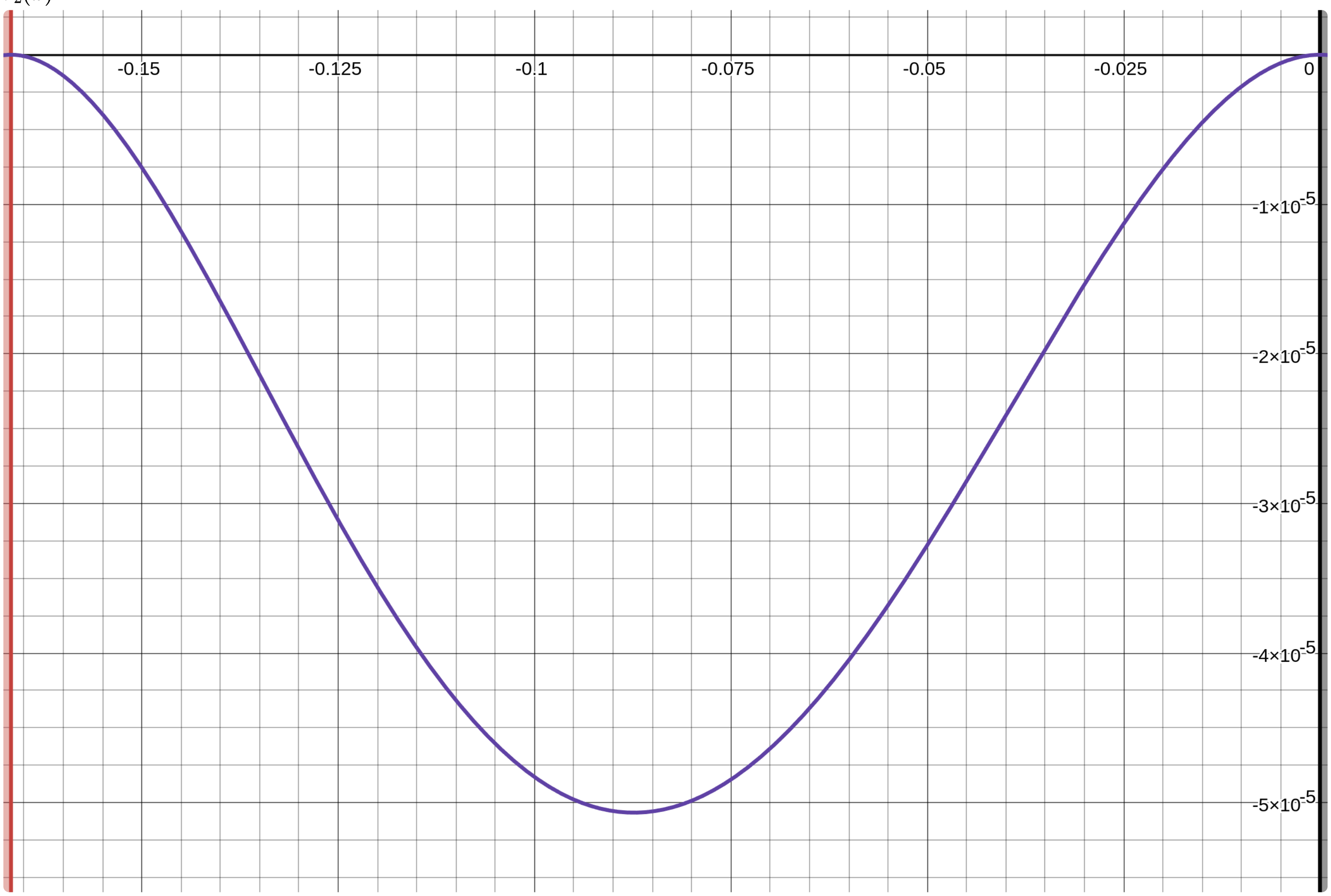
$$H_2(x) = \frac{-(1-3(6x+1)^2+2(6x+1)^3)}{2} + \frac{\pi\sqrt{3}(6x+1-2(6x+1)^2+(6x+1)^3)}{12} + \frac{\pi(-(6x+1)^2+(6x+1)^3)}{6} =$$

$$= x^3(-216+18\pi\sqrt{3}+36\pi) + x^2(-54+3\pi\sqrt{3}+12\pi) + x\pi$$

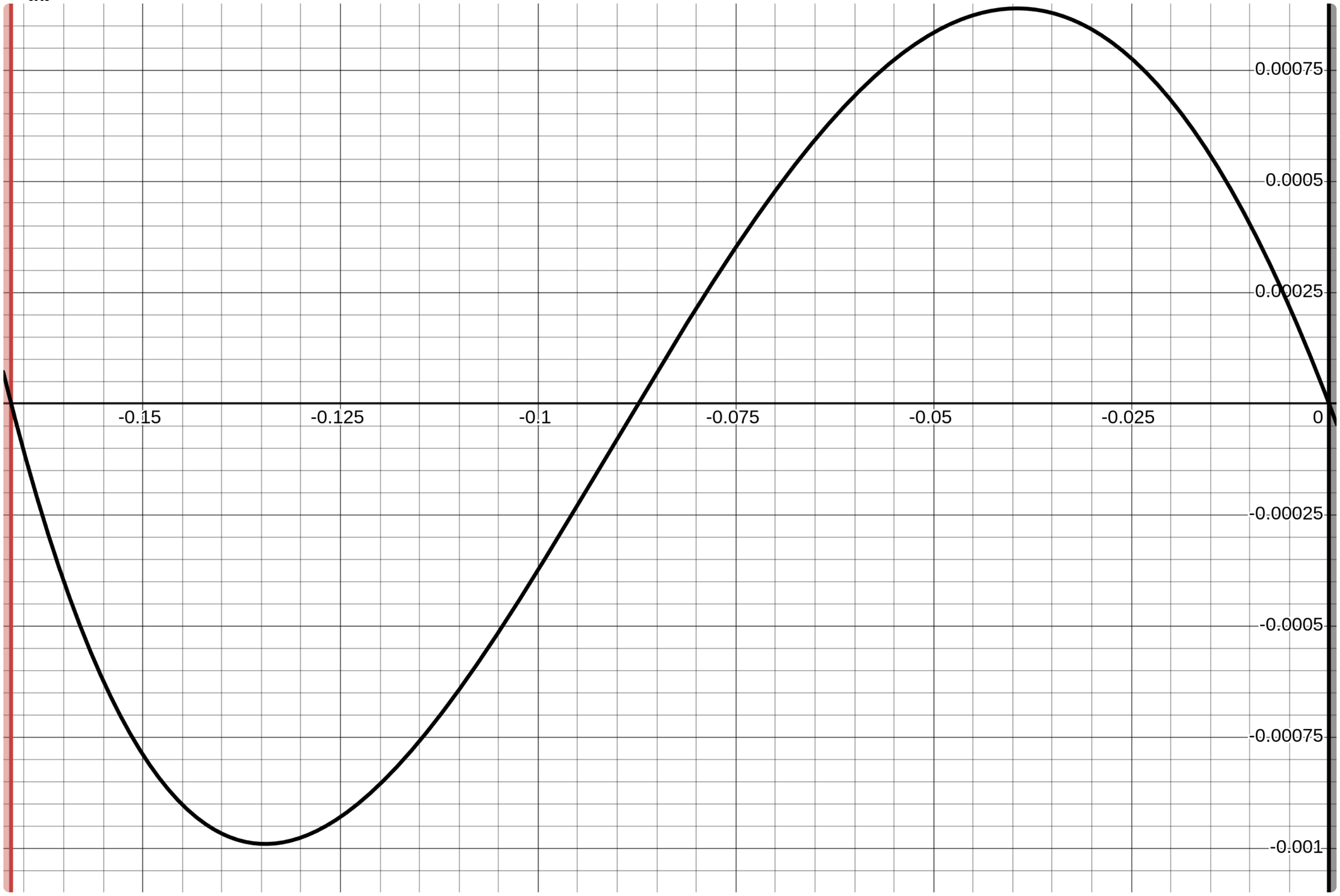
$$r_2(x) = \sin(\pi x) - x^3(-216+18\pi\sqrt{3}+36\pi) + x^2(-54+3\pi\sqrt{3}+12\pi) + x\pi$$

$$\frac{dr_2(x)}{dx} = \pi\cos(\pi x) - 3x^2(-216+18\pi\sqrt{3}+36\pi) + 2x(-54+3\pi\sqrt{3}+12\pi) + \pi$$

$r_2(x)$:



$\frac{dr_2(x)}{dx}$:



$h_3 = \frac{1}{4} - 0 = \frac{1}{4}$; $t_3 = \frac{x-0}{\frac{1}{4}} = 4x$ $\quad x \in [0; \frac{1}{4}]$

$H_3(t) = 0(1-3t^2+2t^3) + \frac{\sqrt{2}}{2}(3t^2-2t^3) + \frac{\pi}{4}(t-2t^2+t^3) + \frac{\pi\sqrt{2}}{2}\frac{1}{4}(-t^2+t^3) =$

$= \frac{3t^2-2t^3}{\sqrt{2}} + \frac{\pi(t-2t^2+t^3)}{4} + \frac{\pi(-t^2+t^3)}{4\sqrt{2}}$

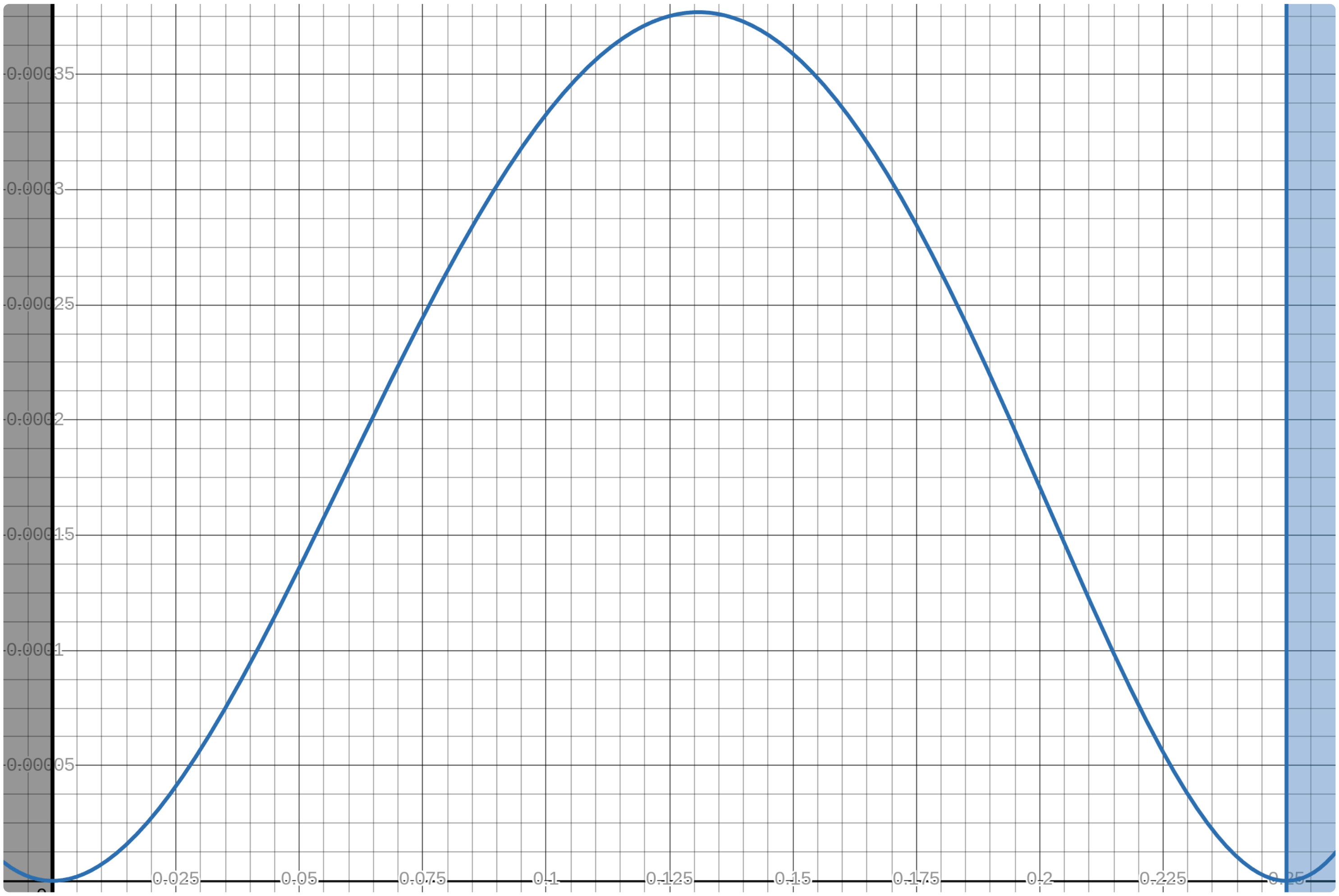
$H_3(x) = \frac{3(4x)^2-2(4x)^3}{\sqrt{2}} + \frac{\pi(4x-2(4x)^2+(4x)^3)}{4} + \frac{\pi(-(4x)^2+(4x)^3)}{4\sqrt{2}} = x^3(-64\sqrt{2}+16\pi+8\pi\sqrt{2}) +$

$x^2(24\sqrt{2}-8\pi-2\pi\sqrt{2}) + \pi x$

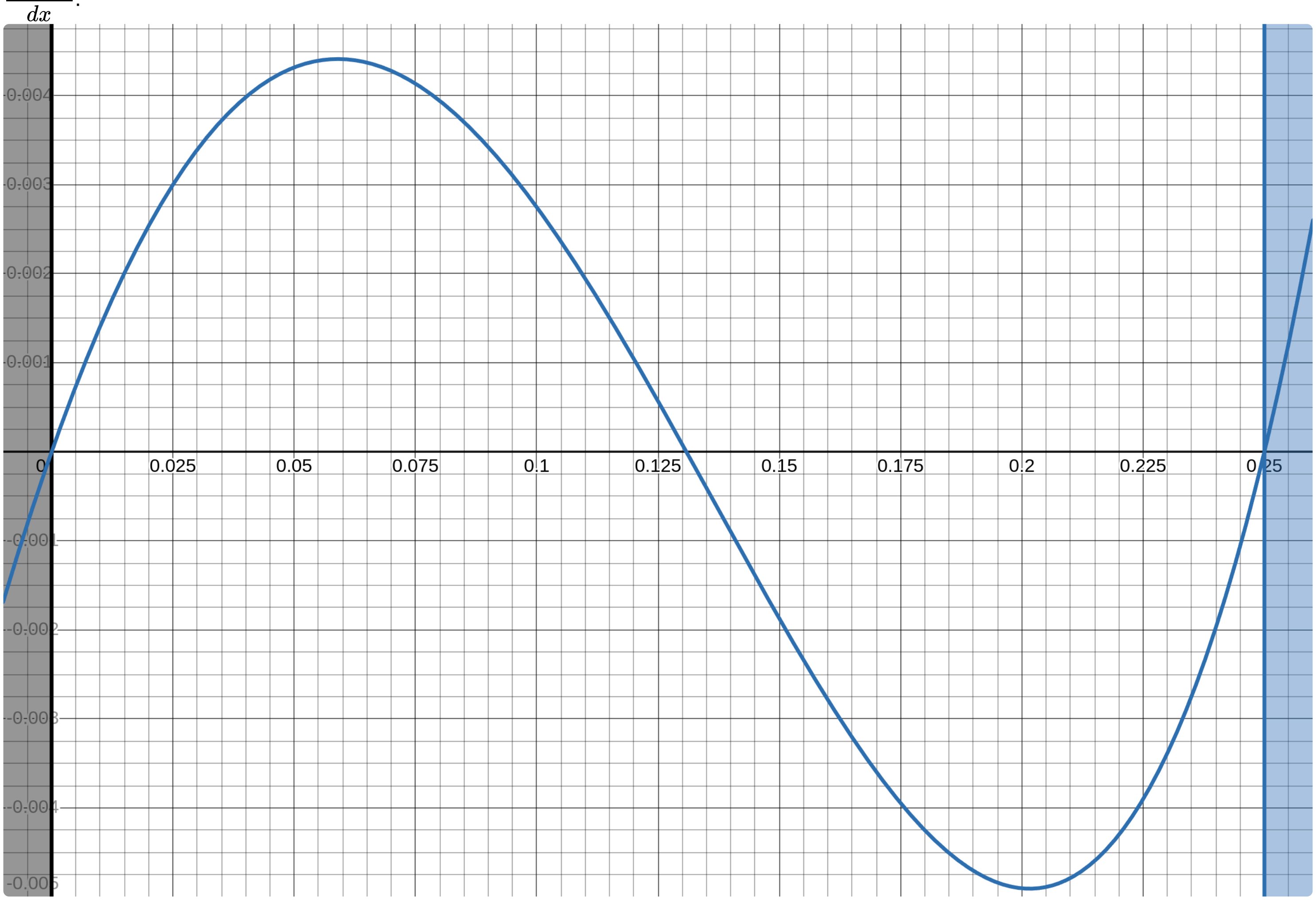
$r_3(x) = \sin(\pi x) - x^3(-64\sqrt{2}+16\pi+8\pi\sqrt{2}) - x^2(24\sqrt{2}-8\pi-2\pi\sqrt{2}) - \pi x$

$\frac{dr_3(x)}{dx} = \pi\cos(\pi x) - 3x^2(-64\sqrt{2}+16\pi+8\pi\sqrt{2}) - 2x(24\sqrt{2}-8\pi-2\pi\sqrt{2}) - \pi$

$r_3(x)$:



$\frac{dr_3(x)}{dx}$:



$H_1(x)$-Синий, $H_2(x)$-Зеленый, $H_3(x)$-Красный, $sin(\pi x)$-Черный